



TEMASEK JUNIOR COLLEGE
PRELIMINARY EXAMINATION
JC2 2021

CANDIDATE NAME

CIVICS GROUP

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H2 BIOLOGY

Paper 2

Structured Questions

9744/02

25 August 2021

2 hours

Candidates answer on the Question Paper

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your name and civics group in the spaces at the top of this page.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's use	
1	/ 9
2	/ 10
3	/ 10
4	/ 10
5	/ 10
6	/ 10
7	/ 10
8	/ 11
9	/ 10
10	/ 5
11	/ 5
Total	/ 100

Answer **all** the questions in this section.

- 1 The tertiary structure in a protein refers to the folding of the protein into a precise, compact, globular three-dimensional shape. This shape is held in place by four types of interactions that occur between R groups of amino acid residues.

(a) State the four types of interactions that hold the tertiary structure in shape. [1]

- **Hydrogen bond, ionic bond, hydrophobic interactions, disulfide bond**

- (b) The Creutzfeldt-Jacob disease (CJD) is a fatal degenerative brain disorder caused by misfolded proteins known as prions. Prions can exist in multiple forms. Examples include the normal prion protein (PrP^{C}) and the disease-causing prion protein (PrP^{Sc}) which causes CJD.

Fig. 1.1 below shows the structures of PrP^{C} and PrP^{Sc} respectively.

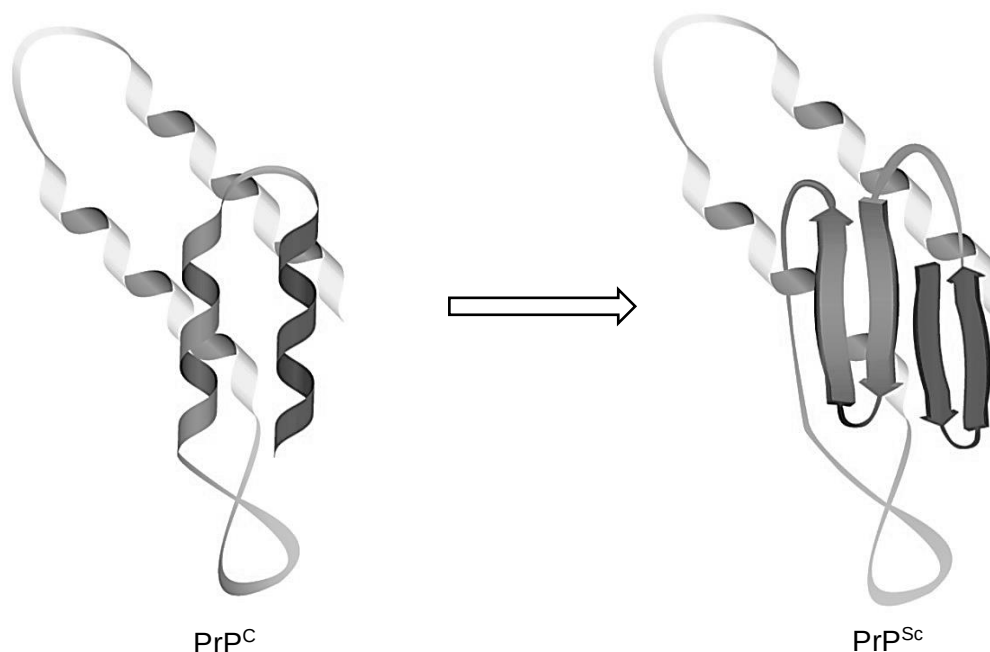


Fig. 1.1

It is noticed that in CJD, each PrP^{C} may misfold into PrP^{Sc} , which then accumulate in large numbers and aggregate (i.e. come together), resulting in the typical sponge-like structure of the brain.

- (i) With reference to Fig. 1.1 and the information given above, suggest how CJD may arise. [2]

1. **When a PrP^{C} misfolds into a PrP^{Sc} , 2 of the α -helices in PrP^{C} are converted into antiparallel β -pleated sheets.**
2. **Hydrogen bonds can form between β -pleated sheets of adjacent PrP^{Sc} proteins, allowing them to aggregate together, resulting in CJD.**

(ii) Suggest why prions challenge the concept of the levels of protein structure. [1]

- **Idea of identical primary structure resulting in different secondary/tertiary structure**

Although CJD is classified as a disease, scientists have noted that there is little to no immune response amongst diagnosed patients.

(iii) Suggest why there is a lack of immune response in patients with CJD. [1]

- **PrP^{Sc}s are produced by own cells and are not recognised as foreign.**

OR

- **PrP^{Sc}s cannot be processed by antigen-presenting cells to stimulate the adaptive immune system.**

(c) (i) Contrast between the structures of PrP^{Sc} and haemoglobin. [2]

Any 2:

feature	PrP ^{Sc}	Haemoglobin
1. Highest level of protein structure	Tertiary	Quaternary
2. Secondary structure	Folded into α -helices and β -pleated sheets	Folded into only α -helices
3. Presence of non-protein component	Only protein component	Protein component + non-protein component (Haem Group)
4. Presence of disulfide bonds	Yes	No

Non-covalent interactions are present in haemoglobin, and this allows for cooperative oxygen binding, thereby increasing the efficiency of oxygen transport.

(ii) Explain how cooperative oxygen binding increases the efficiency of oxygen transport. [2]

1. **The binding of one oxygen molecule to haem group of one (globin) subunit triggers a conformation change in the other three (globin) subunits.**
2. **This increases the affinity of the other (globin) subunits to oxygen / allows for faster uptake of oxygen by subsequent haem groups in the same haemoglobin molecule.**

[Total: 9]

- 2 Fig. 2.1 is an electron micrograph of the process of DNA replication in a eukaryotic cell. The arrows point to replication bubbles. Each replication bubble arises from one origin of replication in the DNA.

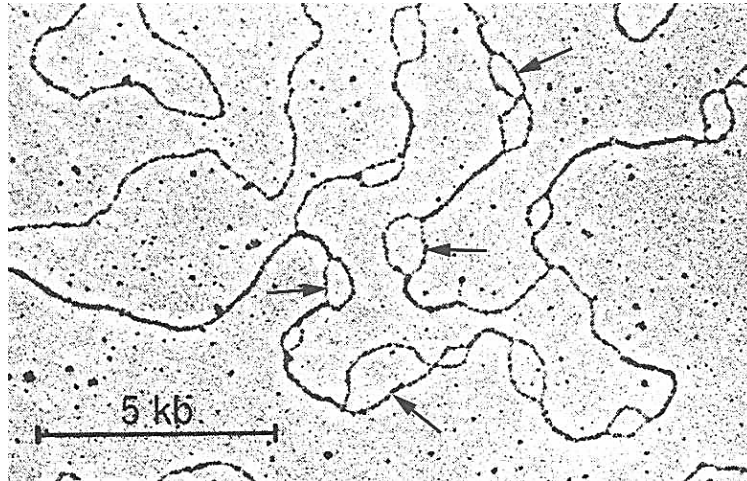


Fig. 2.1

- (a) (i) Explain why eukaryotic cells have multiple origins of replication. [1]
1. Eukaryotic cells have a large amount of DNA to be replicated.
 2. Multiple origins of replication speed up the rate of replication.
- (ii) Explain how the replication bubble is formed and its significance to the semi-conservative mode of DNA replication. [3]
1. Helicase binds to the origin of replication.
 2. This disrupts the hydrogen bonds between complementary bases of double stranded DNA and separates the two DNA strands.
 3. This allows each parental strand to serve as a template for the synthesis of the new DNA strand. Hence, each daughter DNA consists of one parent strand and one new DNA strand.

(b) Fig. 2.2 shows part of a DNA undergoing replication.

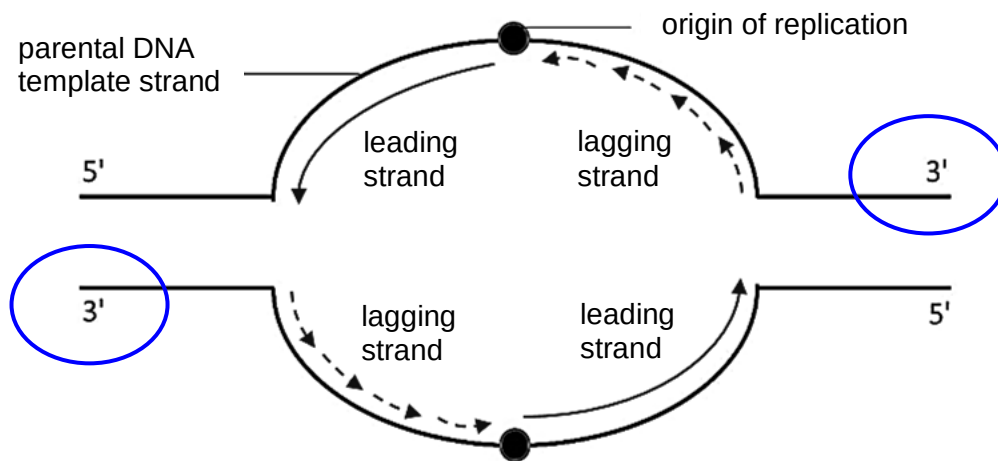


Fig. 2.2

(i) Circle the end(s) of the DNA template(s) on Fig. 2.2 where the end replication problem would be experienced.

[1]

(ii) Explain the reason for the end replication problem. [2]

1. The last RNA primer (bound to the 3' end of the DNA template) that is used to synthesise the last Okazaki fragment of the lagging strand would be removed.
2. DNA polymerase is unable to synthesise / extend DNA from the 5' end of the lagging strand / newly synthesised DNA strand, leaving a portion of the 3' end of the DNA template not replicated.

(c) Contrast DNA replication and polymerase chain reaction. [3]

feature	PCR	DNA replication
1. Separation of double stranded DNA to single stranded DNA	High temperature (95°C)	At lower temperature due to activity of helicase
2. Type of DNA polymerase used	Thermostable DNA polymerase (e.g. Taq DNA polymerase: 72°C)	Normal DNA polymerase (37°C)
3. 3'OH ends for DNA polymerase / type of primers used	Provided by DNA primers	Provided by RNA primers
4. DNA amplified	Small section / specific sequence of DNA is replicated.	Entire set of DNA in the cell is replicated.
5. Synthesis of DNA	New DNA is synthesised in a continuous manner.	Some parts of the new DNA (lagging strand) is synthesised in a discontinuous manner.
6. Temperature of process	Temperature varies. Generally, 3 temperatures 95°C: Denature double stranded DNA 60°C: DNA primers to anneal to DNA template 72°C: Extension of DNA primer	Temperature does not vary / one constant temperature (e.g. 37°C)
7. Maintaining DNA strands as single stranded during synthesis of new DNA	By excess DNA primers	By single-strand DNA binding proteins

[Total: 10]

- 3 The 1968 flu pandemic is one of the deadliest pandemics in history, killing approximately 4 million individuals globally. This pandemic was caused by the novel H3N2 influenza virus.
- (a) Fig. 3.1 shows how the H3N2 influenza virus attacks human respiratory epithelial cells inside the nose.

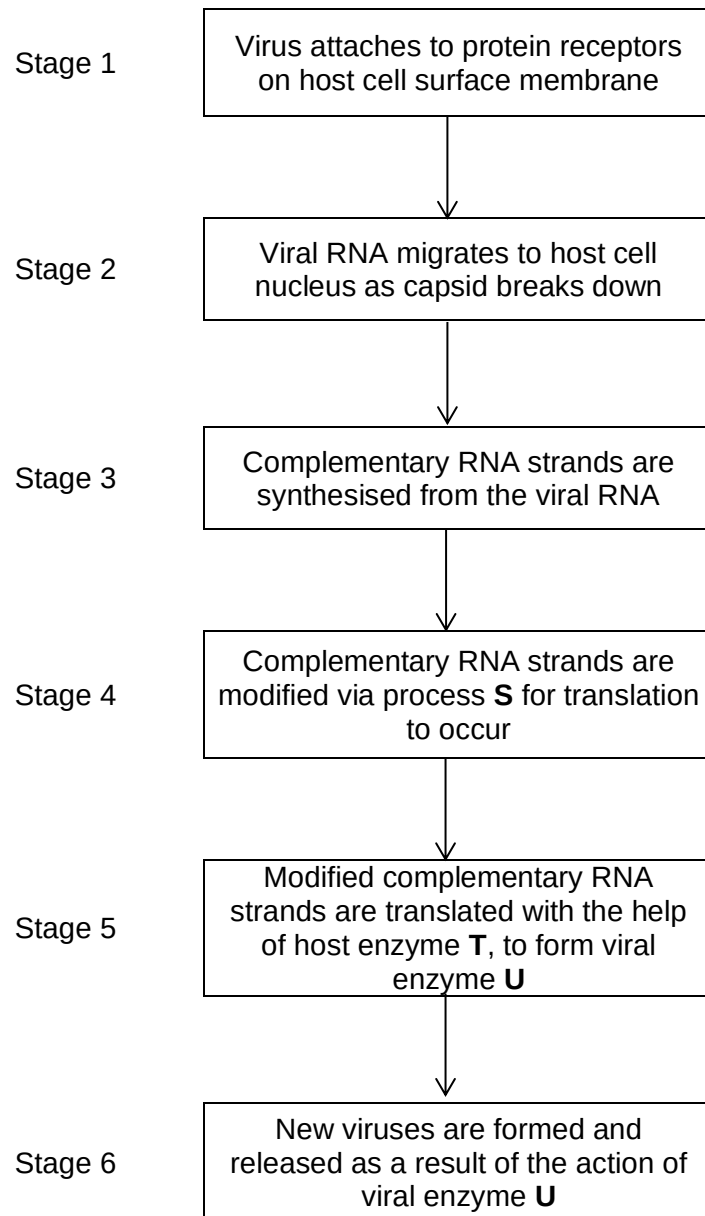


Fig. 3.1

- (i) Explain why the H3N2 influenza virus is only able to attack human respiratory epithelial cells. [1]

1. Haemagglutinin glycoproteins on the viral envelope can only recognise and bind to specific sialic acid receptor proteins on the cell surface membrane of human respiratory epithelial cells.

- (ii) Suggest a specific identity of process S, and explain why process S is required for translation to occur. [2]

1. Process S is the addition of 5' cap.
2. This allows for the small ribosomal subunit to bind for translation initiation to occur.

OR

1. Process S is the addition of 3' poly(A) tail / polyadenylation at 3' end.
1. This protects the complementary RNA from degradation. / This facilitate the export/transport of complementary RNA from the nucleus into cytoplasm for translation to occur.

OR

1. Process S is RNA splicing
2. This allows for the formation of a mRNA molecule with a continuous coding sequence. / This allows different mRNAs to form from the same viral RNA allowing different proteins to be formed.

- (iii) Suggest the identity of host enzyme T, and explain how host enzyme T helps with the translation of the modified complementary RNA strands. [2]

1. Host enzyme T is aminoacyl-tRNA synthetase.
2. It attaches a specific amino acid to the corresponding tRNA.

OR

1. Host enzyme T is peptidyl transferase.
2. It catalyses the formation of a peptide bond between adjacent amino acids.

- (iv) Explain how viral enzyme U helps with the release of new influenza viruses. [1]

1. Viral enzyme U is neuraminidase and it cleaves sialic acid receptors away from the surface of the new influenza particles.

(b) The novel H3N2 influenza virus arose via a mechanism shown in Fig. 3.2.

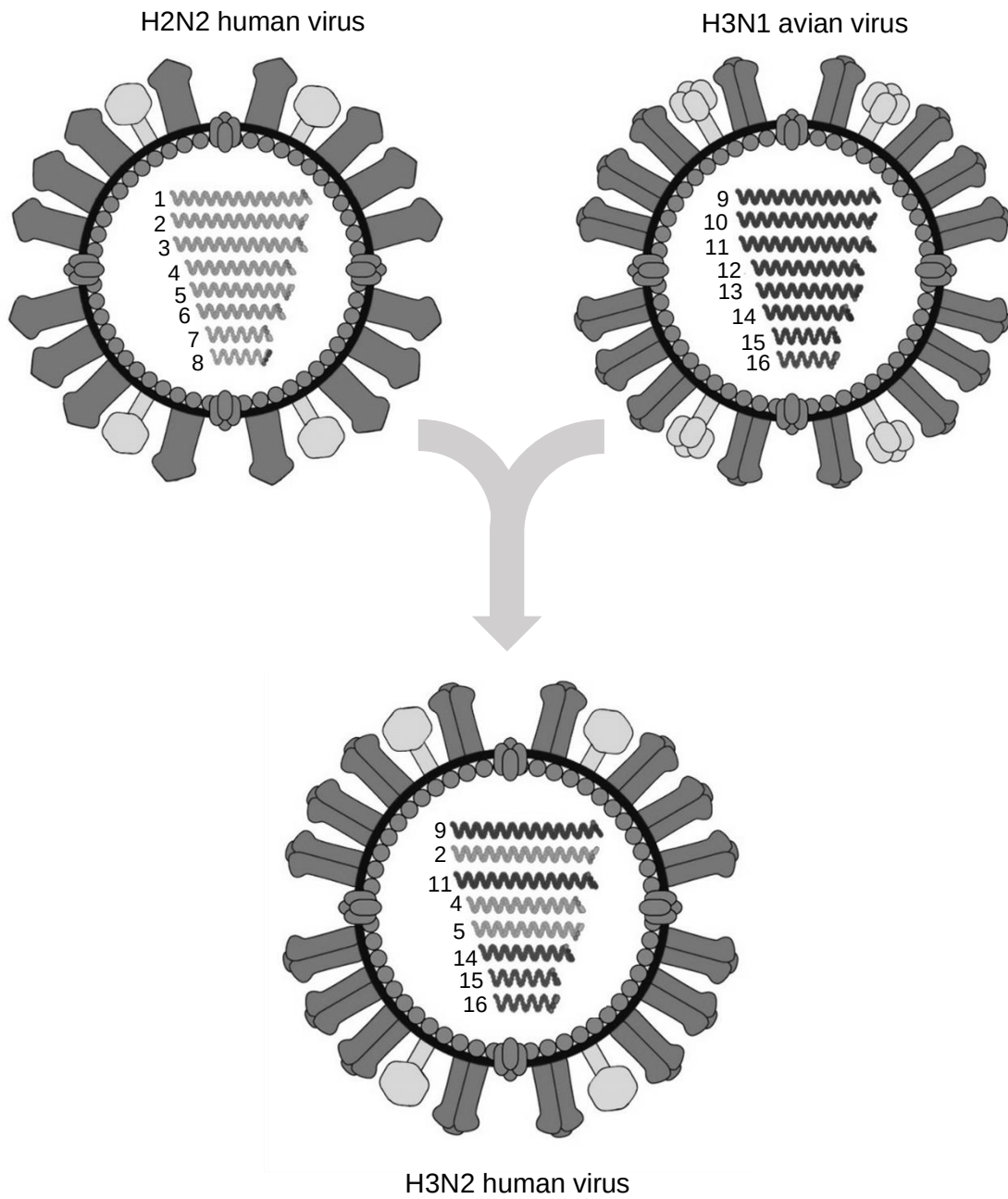


Fig. 3.2

(i) With reference to Fig. 3.2, describe how the novel H3N2 influenza virus arose. [3]

1. Antigenic shift has occurred.
2. Two genetically different strains of influenza virus co-infected the same cell.
3. When the resultant virus was assembled, there was random assembly of the viral RNA segments.
4. RNA segments 2, 4 and 5 were derived from the H2N2 human virus, while RNA segments 9, 11, 14, 15 and 16 were derived from the H3N1 avian virus.
5. This causes the novel virus to have different conformation / structure of surface antigens / haemagglutinin and neuraminidase / glycoproteins.

(ii) Suggest why the human population was susceptible to the novel H3N2 influenza virus. [1]

- Different / new conformation of surface antigens cannot be recognised by pre-existing antibodies / memory cells.

OR

- Different / new conformation of surface antigens can bind more efficiently / effectively to the sialic acid receptors.

[Total: 10]

4 Meiosis, also known as reduction division, is a process by which a parent cell divides to produce gametes.

(a) Assuming that the total number of chromosomes in the parent cell is $2n = 46$, complete Table 4.1.

Table 4.1

stage	number of chromosomes per cell	ploidy	amount of DNA
start of interphase	46	$2n$	$2x$
end of interphase	46	$2n$	$4x$
end of meiosis I (after cytokinesis)	23	n	$2x$
end of meiosis II (after cytokinesis)	23	n	x

[2]

(b) Fig. 4.1 shows the length of kinetochore spindle fibres during different stages of meiosis.

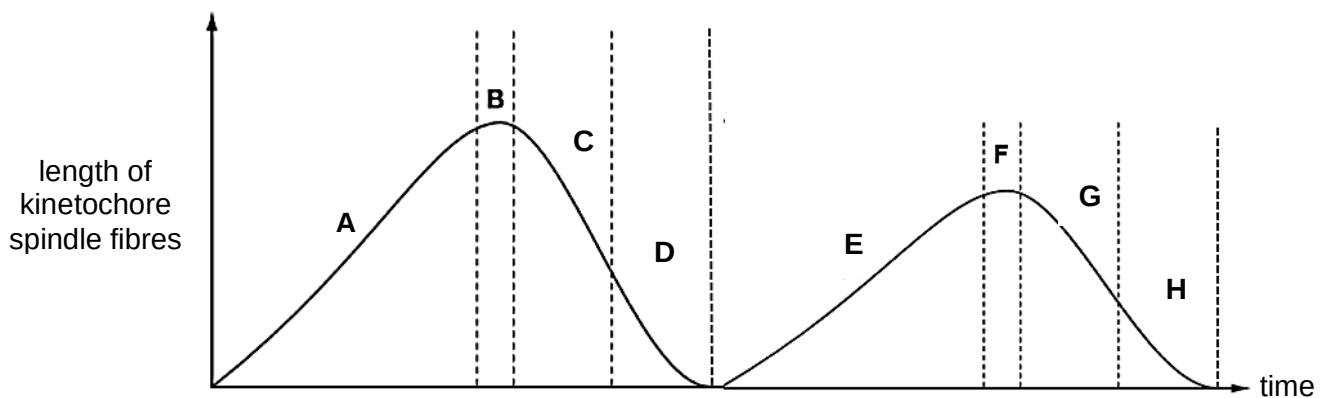


Fig. 4.1

(i) Explain the changes in the length of the kinetochore spindle fibres from A to D. [3]

1. Kinetochore spindle fibres lengthen in A as they extend from each pole towards the equator of the cell
2. Kinetochore spindle fibres are the longest in B as they attach to the kinetochore / centromere of the chromosome lined up at the metaphase plate / equator.
3. Kinetochore spindle fibres shorten in C and D as they pull the homologous chromosomes to opposite poles of the cell.

Fig. 4.2 shows a particular stage in meiosis.



Fig. 4.2

(ii) State the region in Fig. 4.1 that corresponds to the stage shown in Fig. 4.2. [1]

- B or F

(iii) Identify region(s) in Fig. 4.1 that contribute(s) to genetic variation in gametes, and explain how the genetic variation is brought about. [2]

1. A
2. Crossing over between non-sister chromatids of homologous chromosomes during Prophase I leads to new combination of alleles.

OR

1. B and C
2. Independent arrangement of homologous chromosomes at the equator of the cell during Metaphase I and separation of homologous chromosomes during Anaphase I leads to new combination of alleles.

OR

1. F and G
2. Independent arrangement of chromosomes at the equator of the cell during Metaphase II and separation of chromatids during Anaphase II leads to new combination of alleles.

Colchicine is a drug that inhibits spindle fibre assembly. It is commonly used by farmers to induce polyploidy in plants, and this is a common technique used to overcome the sterility of a hybrid species (hybrid sterility) during plant breeding.

For example, triticale is the hybrid of wheat ($2n = 28$) and rye ($2n = 14$). The initial hybrids are sterile as they have an odd number of chromosomes ($2n = 21$). Upon addition of colchicine, the hybrids become fertile, and they can be propagated to become triticale.

(b) Explain how the addition of colchicine overcomes hybrid sterility in triticale. [2]

1. Inhibition of spindle fibre assembly results in the failure of spindle fibres to attach to the chromosomes / chromatids.
2. Sister chromatids are unable to be pulled to opposite poles during anaphase.
3. The resultant daughter cells formed will have double the number of chromosomes.
4. Gametes can now be formed, allowing the hybrids to become fertile.

[Total: 10]

- 5 The majority of lung cancers are caused by long term exposure to several classes of carcinogens present in cigarette smoke. However, there are cases of lung cancers arising in the absence of detectable cigarette exposure.

Analysis of different gene mutations in individuals who smoked ('ever smokers') and those who did not ('never smokers') suggests that lung cancers in never smokers may follow a very different molecular pathway of malignant transformation.

Fig. 5.1 shows the percentage of individuals in US and Asia with lung cancer having mutations in the proto-oncogenes, epidermal growth factor receptor (*EGFR*) gene and *Ras* gene.

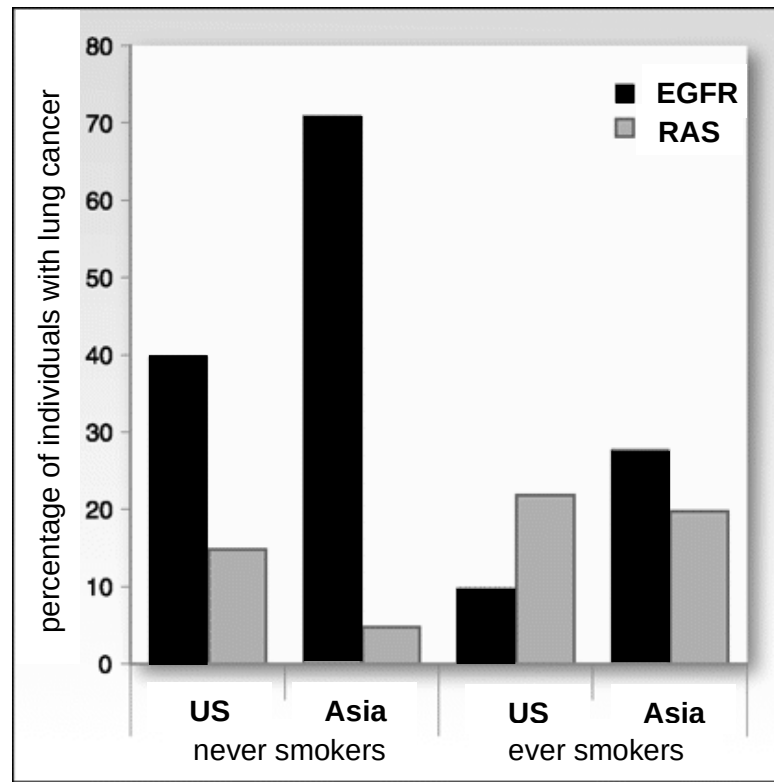


Fig. 5.1

With reference to Fig. 5.1,

(a) (i) Describe the effect of the two gene mutations on the occurrence of cancer in Asians. [2]

1. Mutation in *EGFR* gene has a greater effect in never smokers than ever smokers
2. responsible for 71% of never smokers but only 28% of ever smokers.
3. Mutation in *Ras* gene has a greater effect in ever smokers than never smokers
4. It is responsible for 20% of ever smokers but only 4% of never smokers.

OR

5. For never smokers, *EGFR* gene has a greater effect than *Ras* gene.
6. 71% vs only 4%
7. For ever smoker, *EGFR* gene has a greater effect than *Ras* gene
8. 28% vs 20%

(ii) With reference to Fig. 5.1 and your biological knowledge, explain how never smokers in US and Asia developed lung cancer. [3]

1. Mutation in *EGFR* gene is responsible the majority of lung cancer in never smokers. In both regions, it is responsible for more than 2 fold for never smokers versus ever smokers.
2. Mutation in one copy of *EGFR* gene
3. E.g. constitutively active / resistant to degradation → constantly stimulate cell division
4. **Ref to** gain in function mutation.
5. At the same time, the never smokers also have (loss of function) mutations in their tumor suppressor genes.
6. **Ref to** loss of function mutation.
7. **Ref to** both copies of each tumour suppressor gene are mutated
8. **Idea of** accumulation of gene mutations in a single cell.
9. This results in uncontrolled cell division.

Cancers are traditionally viewed as genetic disorders, but it has become increasingly evident that aberrant epigenetic modifications also play major roles in cancer development. Epigenetics is the study of heritable changes in gene expression that occur without changes in DNA sequences.

Fig 5.2 shows the major epigenetic changes in lung cancer.

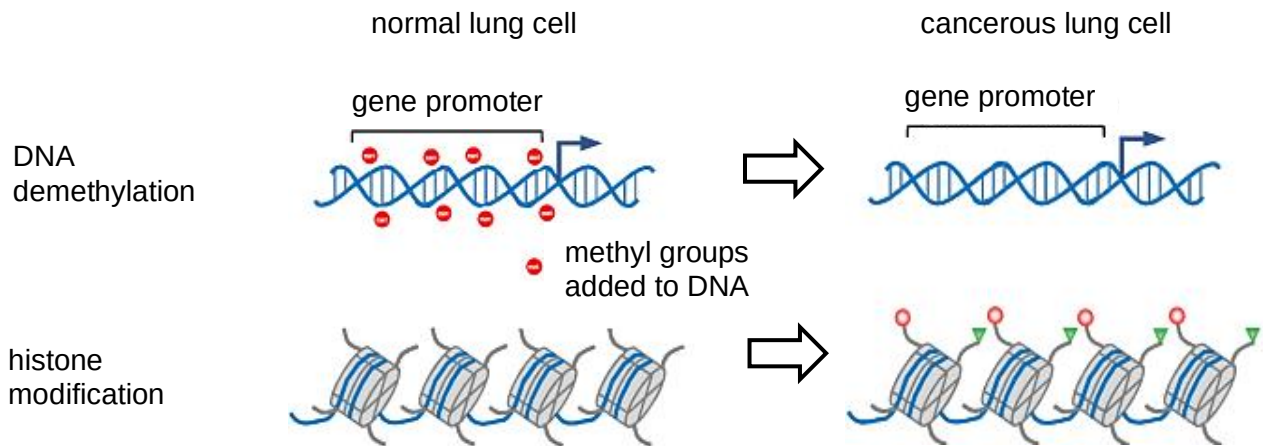


Fig. 5.2

(b) (i) Explain how the epigenetic changes shown in Fig. 5.2 may result in higher gene expression. [4]

1. Removal of methyl groups from the promoter of both genes (proto-oncogenes).
2. Addition of acetyl groups to histones by histone acetylase / acetyltransferase
3. Histones become less positively charged.
4. DNA demethylation and histone acetylation causes DNA to bind less tightly to histones and the DNA (where both genes are found) is less tightly packed, forming euchromatin.
5. Easier for RNA polymerase and general transcription factors to bind at the promoter / transcription initiation complex to form at the promoter
6. Can access promoter of the genes more easily, leading to an increase in transcription.

A non-coding RNA (ncRNA) is a single stranded RNA molecule that is transcribed from DNA but not translated into proteins.

(ii) Suggest how ncRNA may be used as a drug to treat lung cancers arising from epigenetic changes. [1]

1. The ncRNA is complementary to the mRNA produced from the transcription of Ras gene and EGFR gene.
2. It binds to the (oncogenic) mRNA and prevents their translation.

[Total: 10]

- 6 In humans, a gene on chromosome 19 is responsible for the formation of a glycolipid found in red blood cell surface membranes. This glycolipid is known as antigen H. The gene has two alleles, H and h. The product of allele h is non-functional.

A second gene on chromosome 9 codes for an enzyme that modifies antigen H. There are three alleles of this gene:

- I^A which results in the conversion of antigen H to antigen A
- I^B which results in the conversion of antigen H to antigen B
- i which does not produce a functional product so does not modify antigen H.

There are four possible phenotypes depending on the combination of the alleles of these two genes:

- phenotype A, where individuals have antigen A only
- phenotype B, where individuals have antigen B only
- phenotype AB, where individuals have both antigen A and antigen B
- phenotype O, where individuals have neither antigen A nor antigen B.

Two parents, both with phenotype AB, have a child with phenotype O.

- (a) State the genotype of both parents for these **two** genes. [1]

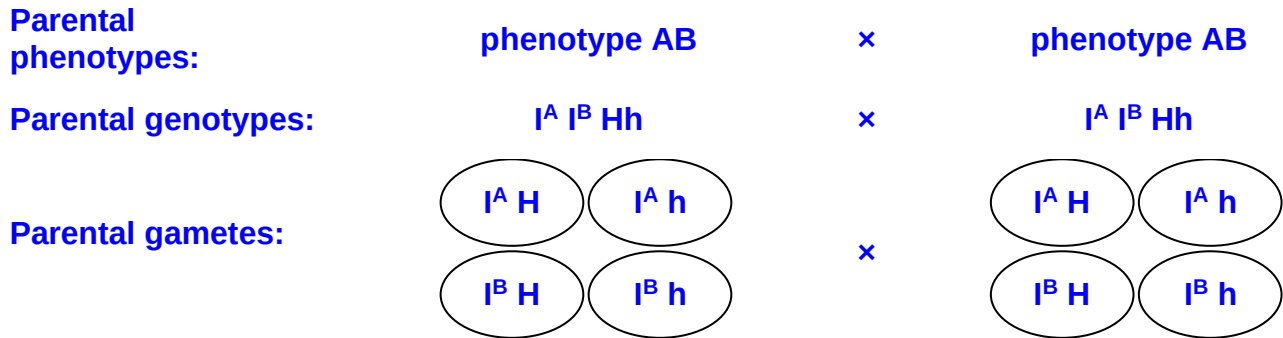
parent 1 $I^A I^B Hh$

parent 2 $I^A I^B Hh$

[1]

(b) The same parents then have a second child.

Use a genetic diagram to show the possible genotypes and corresponding phenotypes of the second child.



Punnett square:

	$I^A H$	$I^A h$	$I^B H$	$I^B h$
$I^A H$	$I^A I^A HH$ phenotype A	$I^A I^A Hh$ phenotype A	$I^A I^B HH$ phenotype AB	$I^A I^B Hh$ phenotype AB
$I^A h$	$I^A I^A Hh$ phenotype A	$I^A I^A hh$ phenotype O	$I^A I^B Hh$ phenotype AB	$I^A I^B hh$ phenotype O
$I^B H$	$I^A I^B HH$ phenotype AB	$I^A I^B Hh$ phenotype AB	$I^B I^B HH$ phenotype B	$I^B I^B Hh$ phenotype B
$I^B h$	$I^A I^B Hh$ phenotype AB	$I^A I^B hh$ phenotype O	$I^B I^B Hh$ phenotype B	$I^B I^B hh$ phenotype O

Offspring phenotype: phenotype A : phenotype B : phenotype AB : phenotype O

Offspring phenotypic ratio: 3 : 3 : 6 : 4

[5]

The interpupillary distance (IPD) is the distance in millimetres between the centres of the pupils of the eyes. Fig. 6.1 shows how IPD is measured.

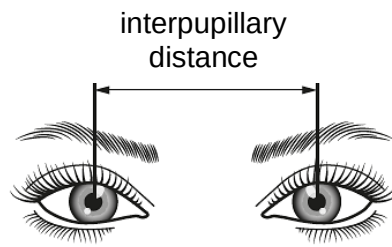


Fig. 6.1

IPD is one example of a characteristic of human facial structure that shows variation.

Fig. 6.2 shows the pattern of variation in IPD in a large sample of adults.

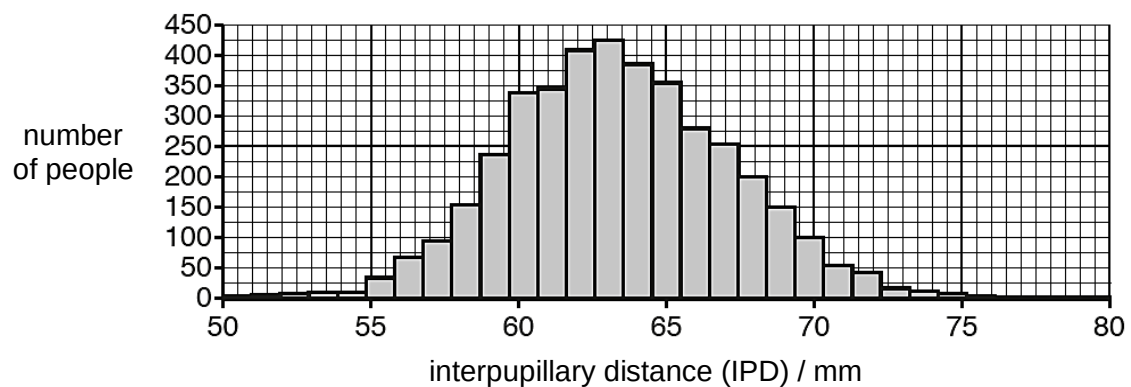


Fig. 6.2

- (a) (i) Name the type of variation shown in Fig. 6.2. [1]

Continuous variation

- (ii) Suggest and explain how genes and the environment contribute to variation in IPD in humans. [3]

1. **IPD is controlled by a number of genes.**
2. **Although the effect of an individual gene is small, the range of phenotypes is a result of the additive / combined effect of various genes.**
3. **Named example of environmental factor affecting gene expression, e.g. age / diet / chemicals / mutagens which affect IPD development**

[Total: 10]

- 7 Cytochrome c is an electron carrier involved in oxidative phosphorylation. It is a protein that transfers electrons between two large proteins found in the electron transport chain. It removes electrons from cytochrome bc1 and transfers them to cytochrome oxidase, as shown in Fig. 7.1.

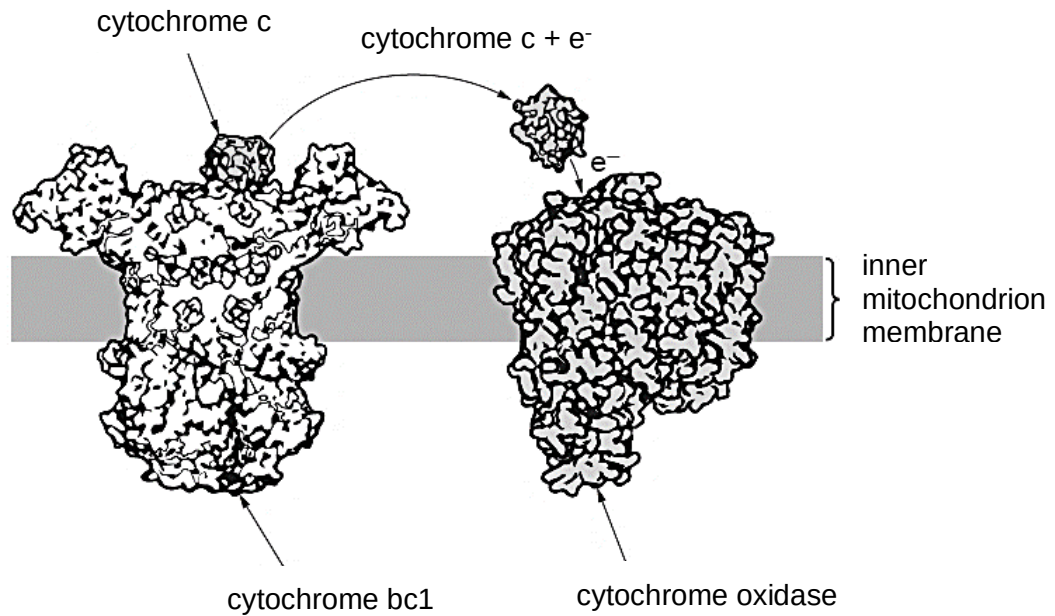


Fig. 7.1

- (a) Cytochrome oxidase is the final protein in the electron transport chain. It is an enzyme which transfers the electrons to an oxygen molecule.

State the final product of this reaction in the mitochondrion. [1]

- Water (H₂O)

- (b) The cytochrome oxidase molecule is different in different species. Fig. 7.2 shows the enzyme from a cow and a bacterium. The shaded section in each represents the polypeptide chains which are very similar in both organisms. This is the area of the enzyme which binds to the cytochrome c.

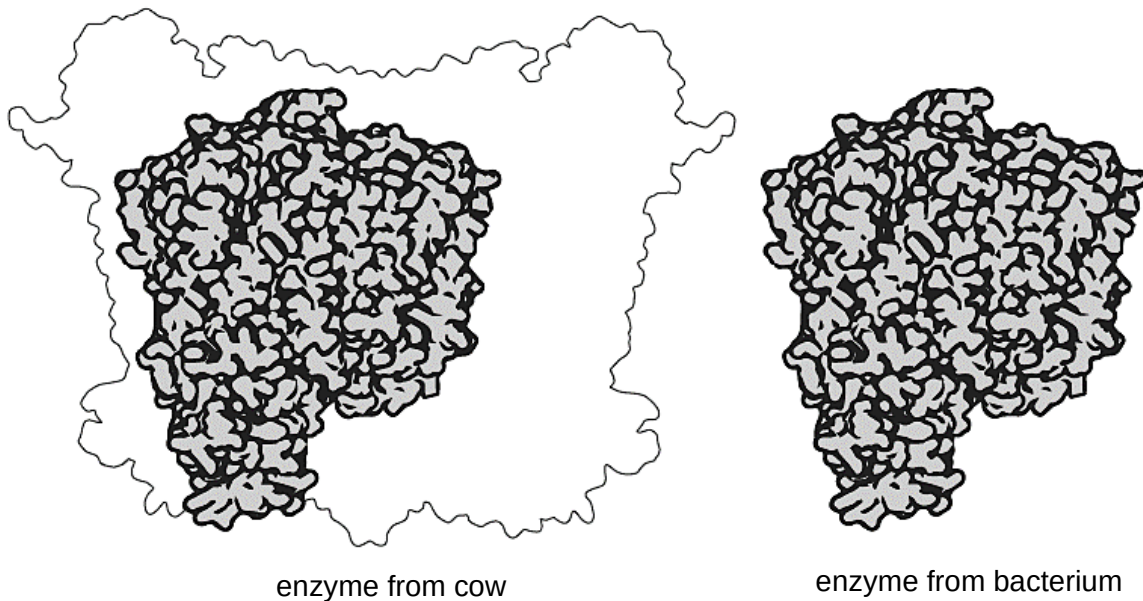


Fig. 7.2

- (iii) Suggest why the shaded regions are similar in all organisms. [1]

1. The DNA sequence that codes for the polypeptide chains is similar in all organisms / has not mutated. Hence, the amino acid sequence / primary structure is similar.

- (iv) Suggest why cytochrome c detaches from the cytochrome oxidase molecule when the electrons it carries are removed. [2]

1. Upon removal of electrons, the shape of cytochrome c is changed.

2. The shape of cytochrome c is no longer complementary to the shape of cytochrome oxidase and does not fit into the active site / cannot form enzyme substrate complexes.

- (c) Cytochrome bc1 is a proton pump that transfers protons into the inter-membrane space.

Antimycin is a chemical that binds to cytochrome bc1 and blocks its action.

Explain what happens to the pH of the inter-membrane space when antimycin is present. [2]

1. **pH increases / becomes less acidic / more alkaline.**
2. **This is because cytochrome bc1 is not able to pump protons into the inter-membrane space, resulting in fewer H⁺ (protons) present.**

- (d) Oligomycin A is another chemical which acts by inhibiting protons from passing through ATP synthase.

- (i) Explain why the presence of oligomycin A in a human causes high levels of reduced NAD to form in the cell. [2]

1. **The proton gradient remains steep as no further protons can be pumped into the inter-membrane space from the matrix of mitochondria.**
2. **Thus, reduced NAD is unable to donate electrons to electron carriers / electron transport chain.**

- (ii) Suggest why lactate begins to accumulate in the blood if high levels of reduced NAD form. [2]

1. **Lactate is the end-product of anaerobic respiration.**
2. **Pyruvate will be reduced to lactate by reduced NAD**
3. **to regenerate small amounts of NAD⁺ for glycolysis to continue to occur.**

[Total: 10]

- 8 An investigation was carried out into the effect of eating two different sources of carbohydrate on the concentration of glucose and the concentration of insulin in the blood plasma. The carbohydrate sources used were glucose and rice. The carbohydrate in rice is mainly in the form of starch.

Volunteers were fed either 50g of glucose or the quantity of rice known to contain 50g of carbohydrate. The concentration of glucose and insulin in their blood plasma was then measured at intervals for 3 hours. The results are shown in Fig. 8.1 and 8.2.

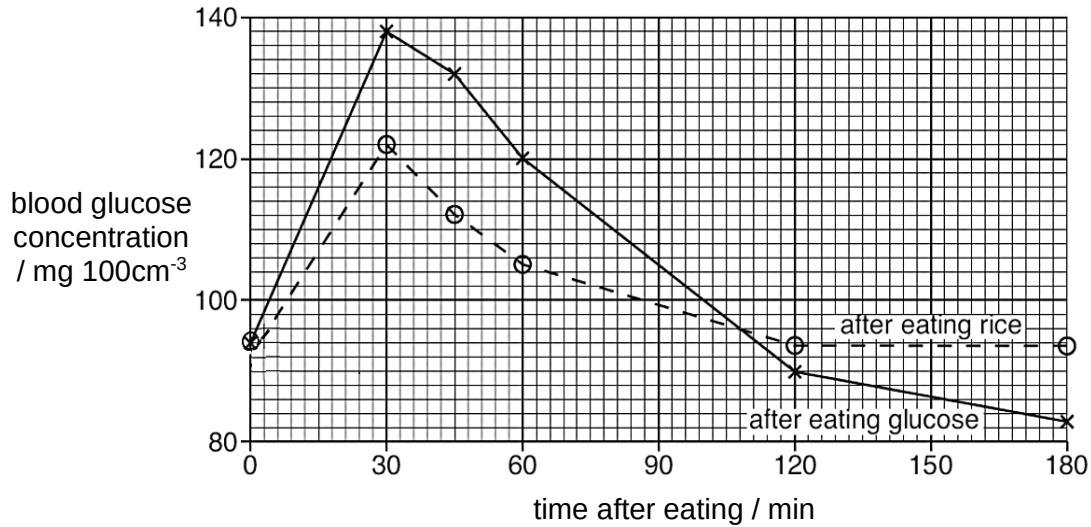


Fig. 8.1

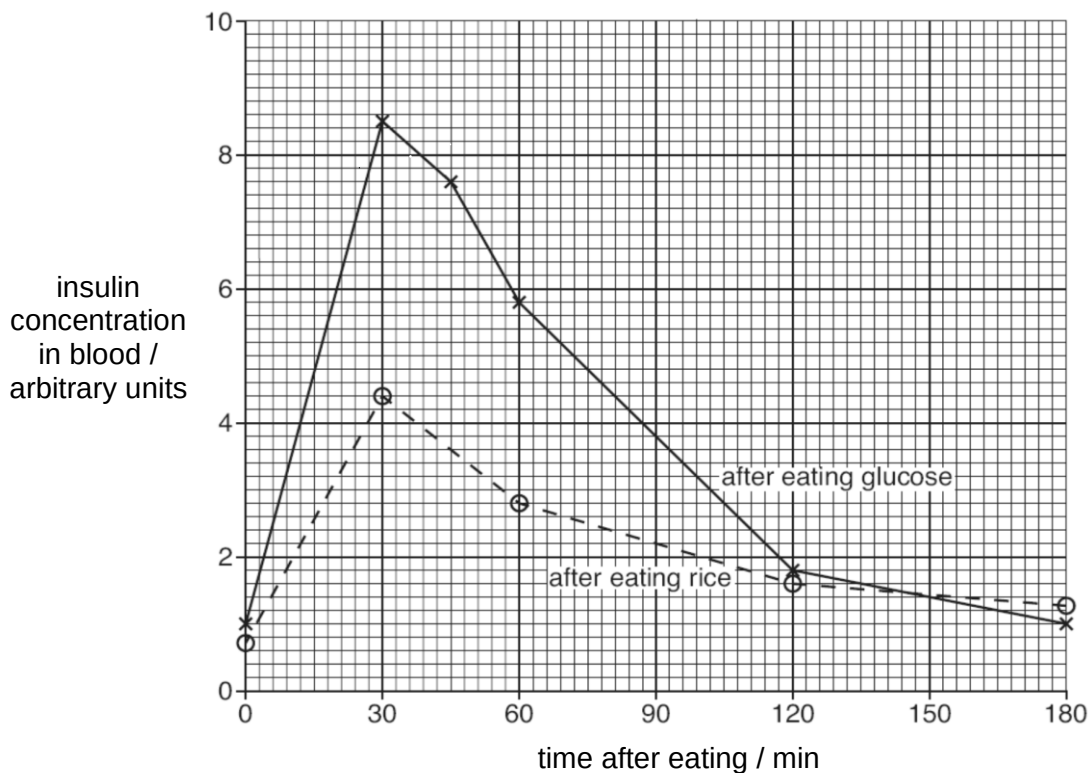


Fig. 8.2

- (a) (i) Calculate the difference between the maximum concentrations of glucose in the blood in the first 30 minutes after eating rice and after eating glucose.

Present your answer as a percentage of the blood glucose concentration at the start of the experiment. Show your working.

$$\begin{aligned} 1. \text{ Difference in maximum concentration} &= 138 \text{ mg100cm}^{-3} - 122 \text{ mg100cm}^{-3} \\ &= 16 \text{ mg100cm}^{-3} \end{aligned}$$

$$2. \text{ Percentage of the blood glucose concentration at the start of the experiment} = (16 \div 94) \times 100 = 17.0\%$$

[1]

- (ii) Give an explanation for the difference you have calculated in (a)(i). [1]

1. Starch needs to be digested to glucose first before it can be absorbed while glucose is quickly absorbed from the start / as it doesn't require further digestion.

- (b) (i) With reference to Fig. 8.1 and Fig. 8.2, explain the difference between the concentration of insulin in the blood in the first 30 minutes after eating glucose and eating rice. [2]

1. Although the concentration of insulin in blood plasma increases with eating both types of carbohydrates, the increase in concentration of insulin for glucose is greater than for rice + QF (8.5 a.u vs 4.4 a.u)

2. This is because the increase in blood glucose concentration after eating glucose is greater than for eating rice. This triggered pancreas to secrete greater amount of insulin into the blood.

- (ii) Explain why the insulin in the blood starts to decrease after 30 minutes. [1]

1. Insulin in the blood starts to breakdown / degrade.

2. As blood glucose concentration decreases towards its normal level, insulin is also no longer secreted.

- (c) Using your biological knowledge on cell signaling, explain how insulin decrease blood glucose concentration. [4]
1. At the target cells, insulin bind to its receptor tyrosine kinase (RTK) resulting in conformational change of the cell surface receptor.
 2. The tyrosine kinase of each subunit phosphorylates the tyrosine residues on (the intracellular tail of) the other subunit (cross-phosphorylation). The receptor is fully activated.
 3. Different relay proteins bind to specific (phosphorylated) tyrosine residues on receptor subunits. The tyrosine kinase of the receptor phosphorylate these relay proteins, which can activate different transduction pathways leading to different cellular responses.
 4. One cellular response is an increase in the amount of glucose transporters (GLUT4) on the cell surface membrane. This increases the membrane permeability to glucose, thus increasing uptake of glucose into the cells from the blood. Hence, blood glucose concentration decreases.
- (d) Explain the advantages of using stem cells to treat patients with end-stage liver disease where their diseased liver cells are unable to carry out normal functions. [2]
1. Stem cells are able to differentiate into liver cells to replace the diseased cells and carry out normal functions.
 2. They also have the ability to undergo long term renewal by mitotic cell division. Thus, there is no need for repeated treatment.

[Total: 11]

- 9 Modern horses are a group of mammals adapted to living in grassland regions. Their evolution can be reconstructed from fossil records.

Table 9.1 shows the evolution of the teeth and height of the modern horse over 50 million years.

Table 9.1

Name	Teeth	Height / cm	Environmental conditions		
<i>Equus</i> (2 mya – today)	“Herbivorous teeth”	150	Predators present.	Grassland region. Mostly fibrous grass.	Edible insects extinct.
<i>Meryohippus</i> (26 mya)	Long cubical molars and flat grinding surfaces	100	Predators present.	Evolution of grass. Forest started to recede.	Edible insects becoming extinct.
<i>Mesohippus</i> (35 mya)	“Omnivorous teeth”	60	Predators present.	Forest filled with trees and shrubs (above 60cm).	Edible insects found on the ground and upper part of the shrubs.
<i>Eohippus</i> (50 mya)	Short, bumpy molars	25	Predators present.	Forest filled with trees and shrubs (above 60cm).	Edible insects found on the ground.

mya - million years ago

- (a) With reference to Table 9.1, explain how natural selection led to the evolution of the horse over time. [5]
1. There were selection pressures in the environment.
 2. Genetic variation was present in the ancestor population due to random mutations.
 3. Predator present and type of food were shrubs and insects.
 4. Early horse ancestors were mostly small / short (25 or 60 cm) and had omnivorous teeth.
 5. Such individuals were able to hide from predators and able to eat insects on the ground or upper part of the shrubs.
 6. They were selected for.
 7. However, environmental conditions changed.
 8. Receding forest, evolution of grass and extinction of edible insects.
 9. Individuals which were taller (e.g. 100 cm or 150 cm) and had herbivorous teeth
 10. were able to escape from predator (e.g. run faster) and derive nutrients from grass.
 11. They were selected for and so survived and reproduced to produce fertile, viable offspring.
 12. Favorable alleles were passed to offspring resulting in higher frequencies of the alleles in the descendent population over time. These led to the modern horses which are tall and have herbivorous teeth.
- (b) The evolution of the horse provides examples of both micro-evolution and macro-evolution. Explain the link between micro-evolution and macro-evolution. [3]
1. Both are driven by the same agents of evolution / natural selection.
 2. *Idea of* both differ on a different time scale and scale of phenotypic changes.
 3. Macro-evolution is the consequence of the effects of accumulated micro-evolution events
 4. *Idea of* micro-evolution is characterised by smaller scale changes in body plans or ways of life that will give rise to the larger scale changes that characterise macro-evolution.
 5. *Idea of* micro-evolution occurring over relatively shorter time spans that forms the long time span that characterise macro-evolution.

Horses have hooved feet and an odd number of toes on each foot. This feature supports the common ancestry of horses and tapirs. Unlike the horse, the tapirs retained their jungle specialisation.

- (c) Suggest how this difference led to the speciation of their common ancestors into modern horses and tapirs starting from about 26 mya. [2]
1. From 26 mya, the horse ancestors started living outside the forest but the tapir ancestors remained in the forest. This led to disruption of gene flow between the two groups of mammals / prevents interbreeding.
 2. Forest and grassland had different environmental conditions / different selection pressures. Hence, natural selection operated differently. The two groups evolved independently from each other over time and became different species.

[Total: 10]

10 Countries that have a high number of cases of malaria also have problems with diseases caused by bacteria. This means that many people in these countries are prescribed antibiotics, such as penicillin, for the treatment of bacterial infections.

(a) Outline how penicillin acts on bacterial cells. [2]

- 1. Penicillin binds irreversibly to the active site of transpeptidase, thus inhibiting the peptide cross-linking of two peptidoglycan chains.**
- 2. Therefore, the cell wall of dividing bacterium becomes weaker. Osmotic lysis occurs.**

The female *Anopheles* mosquito is the vector of the *Plasmodium* (a type of unicellular eukaryotic organism) pathogen that causes malaria. The insect takes in *Plasmodium* when taking a blood meal from an infected person. At a later stage, the insect can transmit the pathogen when taking a blood meal from an uninfected person.

(b) Research has shown that *Plasmodium* is not always transmitted to uninfected people.

Two main reasons for this have been suggested.

- The immune system of mosquitoes kills *Plasmodium* while it is still in their gut.
- Bacteria living in the gut of mosquitoes compete with *Plasmodium* so it does not survive to continue its life cycle.

A study was carried out to see if taking antibiotics affects the risk of transmission of malaria.

Some observations and results of the study are summarised in Fig. 10.1.

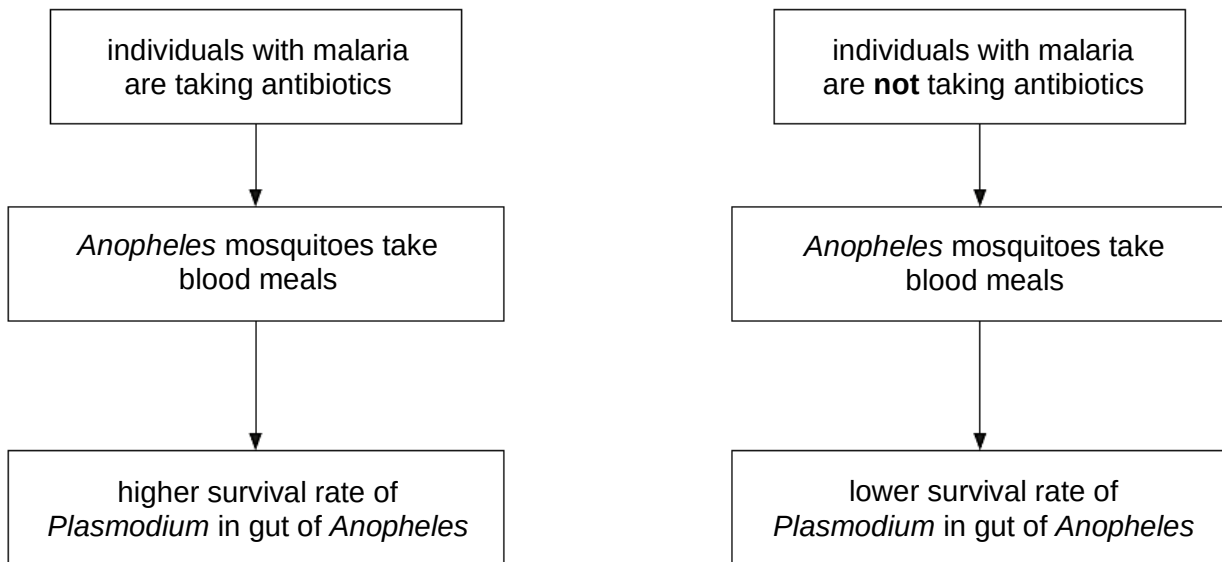


Fig. 10.1

Suggest explanations for the results shown in Fig. 10.1 and comment on the importance of these results for doctors working in countries that have malaria and a high number of bacterial infections. [3]

1. Use of antibiotics may increase incidence / number of cases of malaria.
2. People taking antibiotics for bacterial diseases will pass on antibiotics to *Anopheles* when it feeds, so antibiotics kill gut bacteria in *Anopheles*.
3. Hence, there is decreased / no competition between gut bacteria and *Plasmodium*, leading to higher survival of *Plasmodium* which makes it difficult for an effective immune response to be mounted against *Plasmodium*.

Any 1:

4. There is a need to balance antibiotic intake with increased malaria transmission.
5. Treat for malaria before giving antibiotics (for non-serious / non-life threatening) bacterial infections.
6. Only prescribe antibiotics that have no / low impact on bacteria in *Anopheles* gut.
7. Avoiding use of antibiotics to treat malaria.
8. AVP (need to research which antibiotics have this effect / alternative to antibiotics to treat bacterial infections)

[Total: 5]

11 Fig. 11.1 shows the percentage of a population with HIV and the number of new tuberculosis (TB) infections from 1980 to 2010.

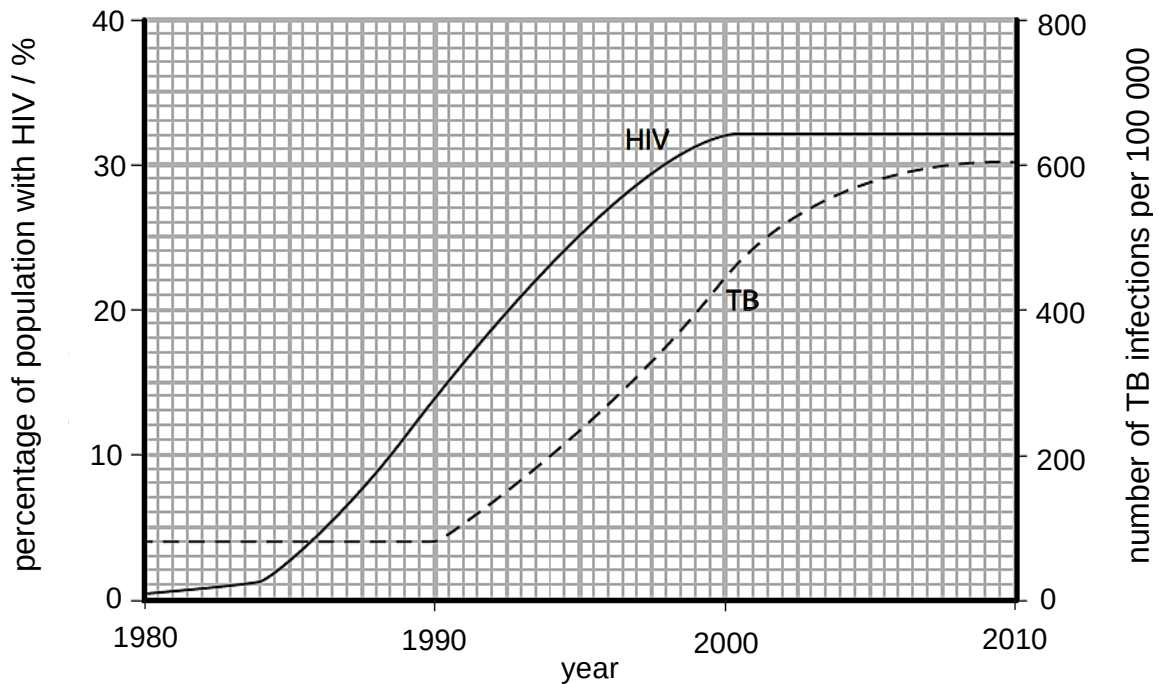


Fig. 11.1

(a) Describe the evidence shown in this graph that suggests a relationship between infection with HIV and TB infection. [2]

1. There is no indication of any relationship between HIV and TB infection before 1990 / only indication of a relationship after 1990.
2. From 1990 to 2010, the % of population with HIV increases from 14 to 32. This increase is also similar / corresponded to a similar increase in number of TB infections per 100 000 people from 80 to 440.
3. Also, the increase in TB infections comes after an increase in HIV infections.

TB is caused when droplets, containing the bacterium *Mycobacterium tuberculosis*, are inhaled into the lungs. In the lungs, large numbers of the bacterium are formed rapidly. These can be ingested by macrophages. Eventually, tubercles (tissue masses), containing dormant bacteria inside macrophages, may form.

(b) Describe the events from the ingestion of *M. tuberculosis* to the formation of tubercles. [3]

1. Inside phagosome, *M. tuberculosis* inhibits fusion of phagosome with lysosome, hence phagolysosome cannot be formed and no lysosomal enzymes available to kill the bacteria.
2. *M. tuberculosis* then survives and reproduce within the macrophages by inhibiting lysosomal enzymes / having the ability to escape out of the phagosomes into the cytosol.
3. The infected macrophages releases cytokines, which recruits more macrophages and other immune cells to the infected alveoli, forming tubercles.

[Total: 5]